EVERY DIFFERENTIAL GRADED MODULE ADMITS A PROPERTY (I) CORE RESOLUTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the inductive co-resolution through lemma-good-sub, the product-total module with its explicit grading and differential, the interleaved filtration F_{2i} and F_{2i+1} with products of shifts of A-dual as quotients, the admissibility of the inclusions through lemma-source-graded-injective, and the quasi-isomorphism of the augmentation are all retained verbatim.

Editorial note. The label lemma-right-resolution names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Good sub-embeddings of differential graded modules. Same chapter, Lemma lemma-good-sub. Supplies, at each induction step, the embedding M_i to I_i whose cokernel continues the resolution and whose two-term structure produces the graded pieces.
- Injectivity of products of shifted duals and the right-resolution spectral sequence. Same chapter, Lemma lemma-source-graded-injective; Homology, Lemma homology-lemma-good-right-resolution-gives-qis. Shows the filtration inclusions are admissible monomorphisms, and turns the constructed co-resolution into the quasi-isomorphism M to I.

Source reference. Original lemma 09J0: sources/stacks-chow/tags.txt, source lines 12058–12058. Prerequisite pointer invoked at proof line(s) 2751; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09KT: sources/stacks-dga/pinned-original-source.tex, source lines 2655–2655. Prerequisite pointer invoked at proof line(s) 2723; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09K2: sources/stacks-dga/pinned-original-source.tex, source lines 2317–2317. Prerequisite pointer invoked at proof line(s) 2747; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules such that*

(1) *$M \to I$ is a quasi-isomorphism, and*
(2) *I has property (I).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_i \to I_i \to M_{i+1} \to 0$$

where the maps $M_i \to I_i$ are chosen as in Lemma 09KT. This gives a "resolution"

$$0 \to M \to I_0 \xrightarrow{f_0} I_1 \xrightarrow{f_1} I_1 \to \ldots$$

Denote I the differential graded A-module with graded parts

$$I^n = \prod\nolimits_{i \geq 0} I_i^{n-i}$$

and differential defined by

$$\mathrm{d}_I(x) = f_i(x) + (-1)^i \mathrm{d}_{I_i}(x)$$

for $x \in I_i^{n-i}$. With these conventions I is indeed a differential graded A-module. Recalling that each I_i has a two step filtration $0 \to I_i' \to I_i \to I_i'' \to 0$ we set

$$F_{2i} I^n = \prod\nolimits_{j \geq i} I_j^{n-j} \subset \prod\nolimits_{i \geq 0} I_i^{n-i} = I^n$$

and we add a factor I_{i+1}' to $F_{2i} I$ to get $F_{2i+1} I$. These are differential graded submodules and the successive quotients are products of shifts of $A^\vee$. By Lemma 09K2 we see that the inclusions $F_{i+1} I \to F_i I$ are admissible monomorphisms. Finally, we have to show that the map $M \to I$ (given by the augmentation $M \to I_0$) is a quasi-isomorphism. This follows from Homology, Lemma 09J0. □